

Advancing Frontiers of Energy Forecasting with Novel Methods

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What does it cost to be confidently wrong?

- Electricity cannot be stored at grid scale. Supply and demand must match every second
- Someone must guess tomorrow's demand today
- Guess low and there isn't enough power. Guess high and everyone pays for generators that idled
- February 2021: forecasts missed, reserves fell short, and ERCOT cut power to homes that were paying for it
- Operators can hedge and start generators early, before the market says it's economical. That insurance costs money, and it lands on customers' bills



Texas during Winter Storm Uri (2021)
Photo source: Houston Chronicle (2021)

Data Collected



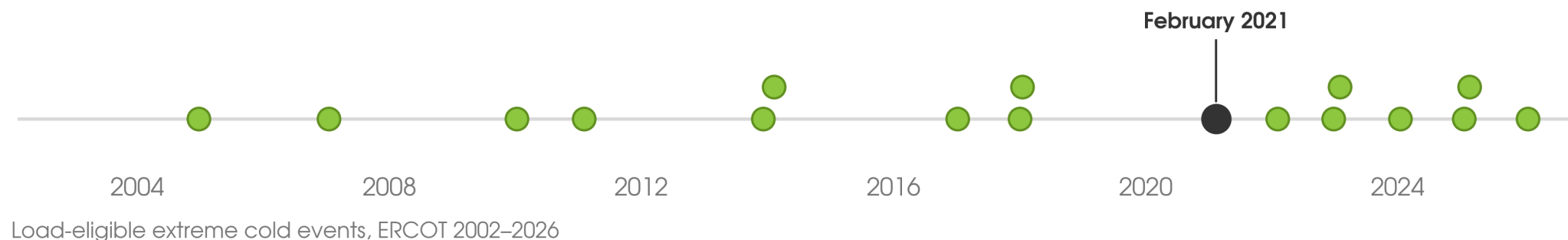
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How many extreme cold events are in twenty-four years of Texas grid data?

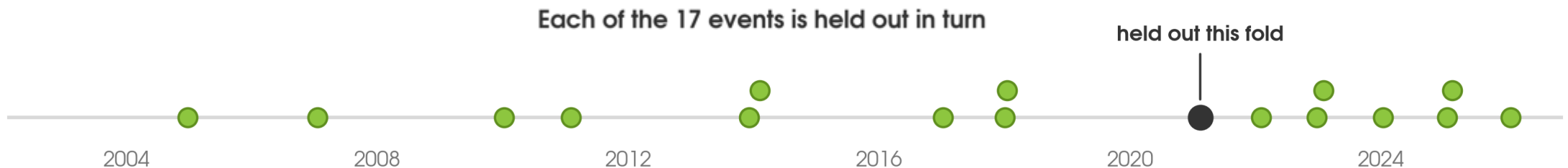
- Hourly ERCOT system load, 2002-2026 adopted by content hash, so provenance is exact
- A model can learn from ten thousand ordinary days and has almost nothing to learn from for the days that decide whether the grid holds
- Worse, the ordinary days actively mislead: the relationship between weather and demand changes in extremes
- Of 1,271 event-hours: 80 documented as load-shed, 0 documented as not shed, 1,191 undetermined

Seventeen extreme cold events in twenty-four years



How do you learn from seventeen examples?

- AdaCNP (Adaptive Conditional Neural Processes) predicts by referring to similar past days, weighting past freezes more heavily than past Tuesdays
- Published in 2026 and demonstrated on the PJM & ISO-NE grids. I asked whether it holds on ERCOT
- Compared against a standard CNP baseline, the weighting is the single intended difference
- Pre-registered: the endpoint, the test, and the decision rule were committed to version control before any confirmatory run



Nothing within seven days of the held-out event may inform it

One fold, schematic — not to scale

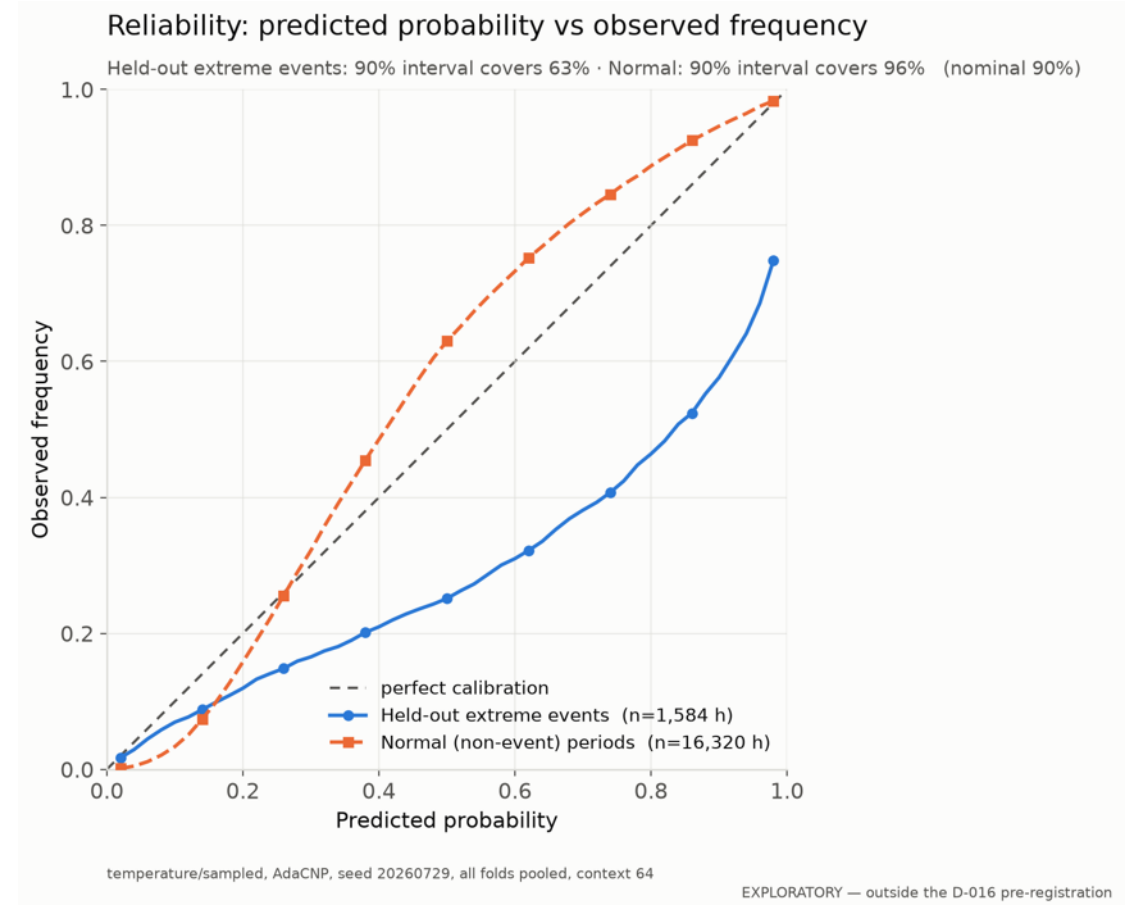


Research Findings



A 90% interval that is right 63% of the time

- A 90% interval covers 96% of normal hours, but only 63% of held-out extreme-cold hours
- The failure is directional: the models under-forecast demand while reporting high confidence
- It gets worse as events get worse, roughly 3.5 points of coverage lost per °C of cold margin
- The replication failed at first, and the reason mattered: the model couldn't see temperature, while the events themselves are defined by temperature



Research Findings

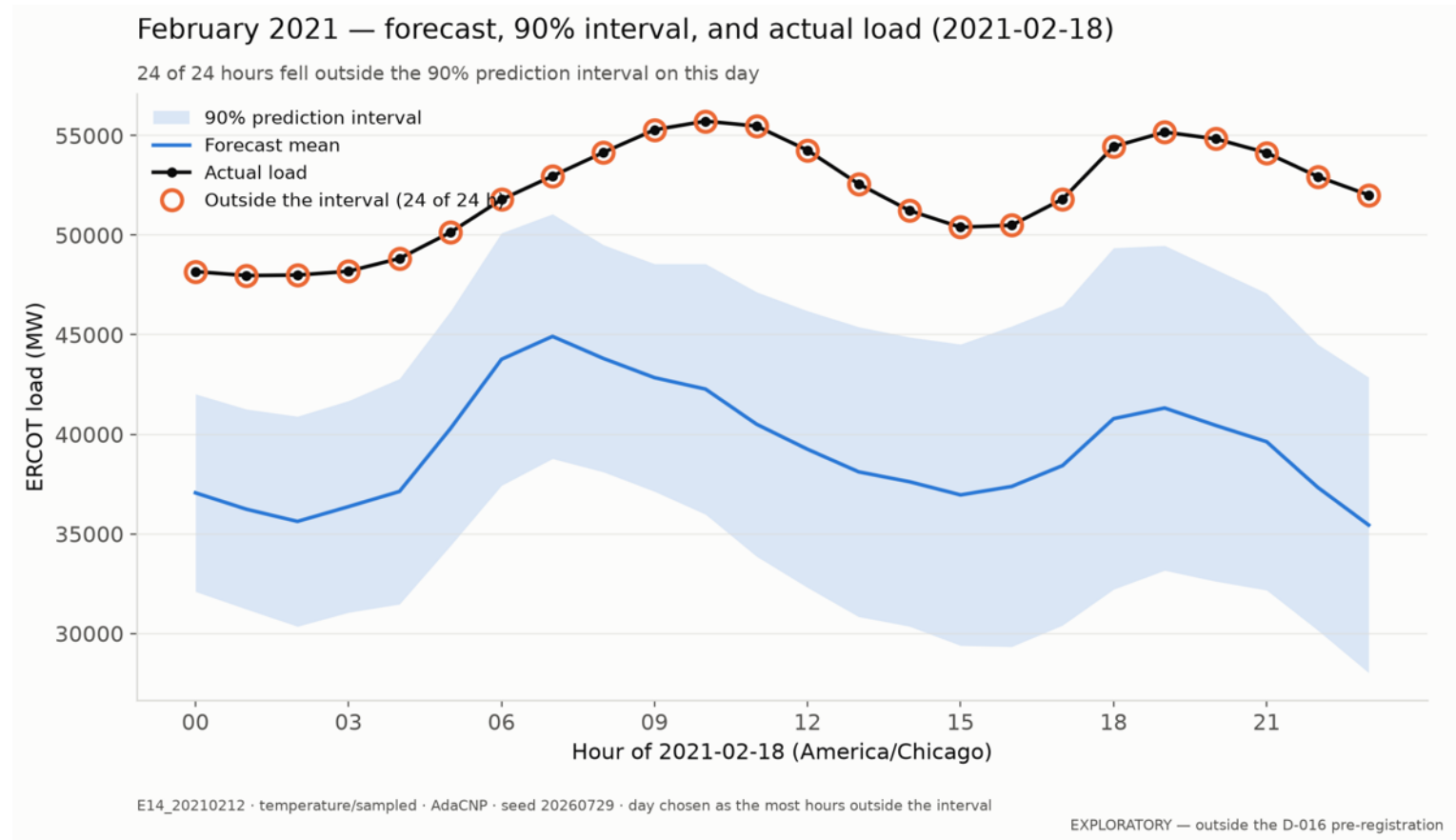


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February 18, 2021: Every hour outside the interval

- All 24 of 24 hours fell outside the 90% prediction interval
- Observed demand ran roughly 10 GW above the forecast mean, all day
- No load shedding occurred on this day. This is forecast error, not curtailed demand
- The worst day of that event, selected by a stated rule rather than chosen after the fact



The method helped at the edge of what seventeen events can resolve

- Pre-registered test: AdaCNP beat the standard CNP baseline by +0.459 in the primary score, 95% CI [+0.108, +0.811], $p = 0.014$, across 17 events
- Robustness holds: Wilcoxon $p = 0.020$, the sign survives leaving out any single event, and 11 of 17 events favor AdaCNP
- But the effect sits just below the minimum detectable effect of 0.495, about 77% power at the size found
- But it appears only in the pre-registered configuration. Nearest-neighbor retrieval favored the baseline, at -0.240
- But it is not the effect size Hu et al. reported. Detecting that would need several hundred events; Texas has seventeen

A wrong forecast is survivable. A confidently wrong one is not.

- Confidently wrong is what makes an operator skip an expensive hedge, the insurance that February 2021 shows is worth buying
- The unreliability is predictable. Calibration degrades measurably as events get more severe
- A model can be built to flag when its own interval should not be trusted
- For an early-commitment decision, knowing when not to trust the forecast is worth more than a slightly better forecast
- Research priority: fix calibration before changing architecture

What these models get wrong matters more than which one won

- The method helps modestly, and near the limit of what seventeen events can resolve
- Both models are badly and directionally overconfident exactly when the grid is under stress
- The pre-registration and the reported power analysis are what make both of those statements admissible
- Future work: acquire day-ahead forecast weather, broaden the event definition to raise the sample size, add seeds

Acknowledgements



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References: Hu, C., Ma, Y., Wu, Y., and Hou, Y. "Resilient Load Forecasting under Climate Change: Adaptive Conditional Neural Processes for Few-Shot Extreme Load Forecasting."
arXiv:2602.04609, 2026.

Thank you!

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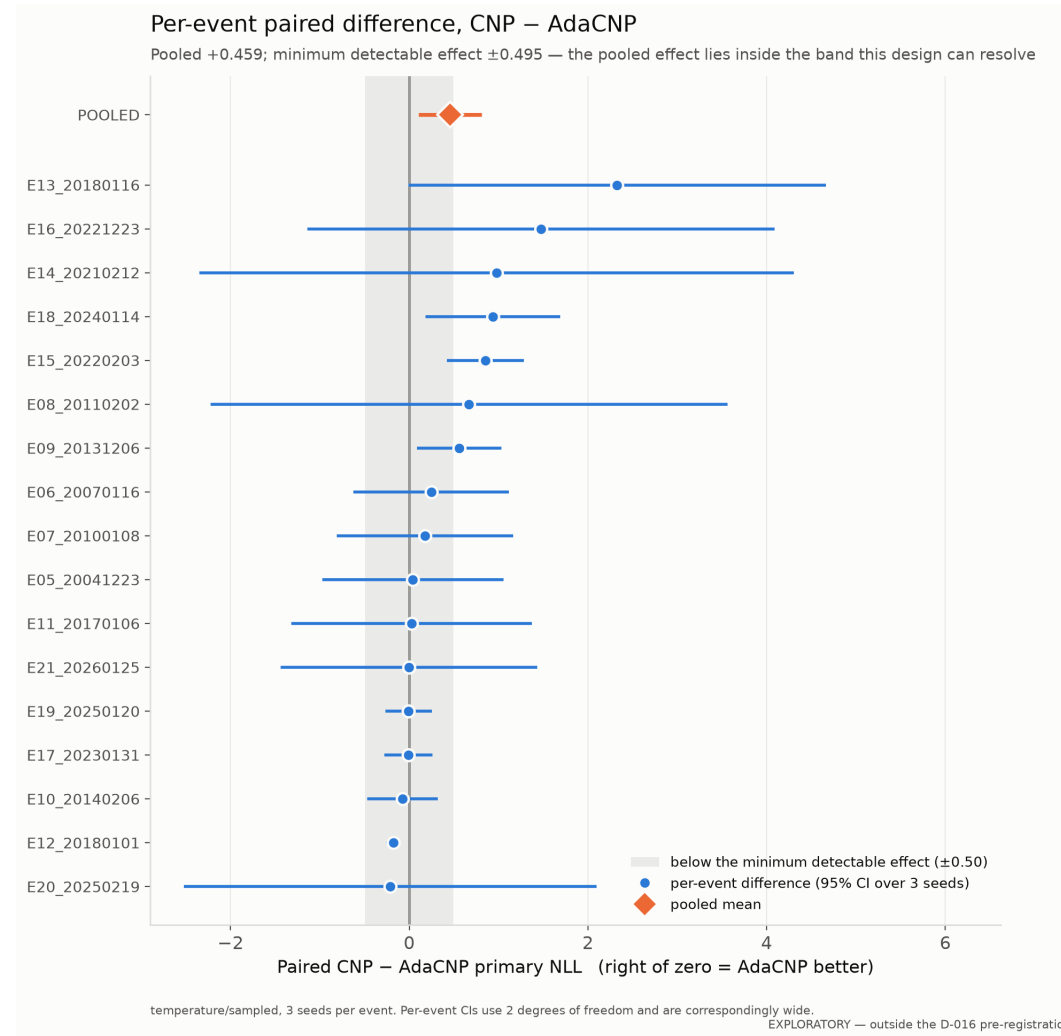


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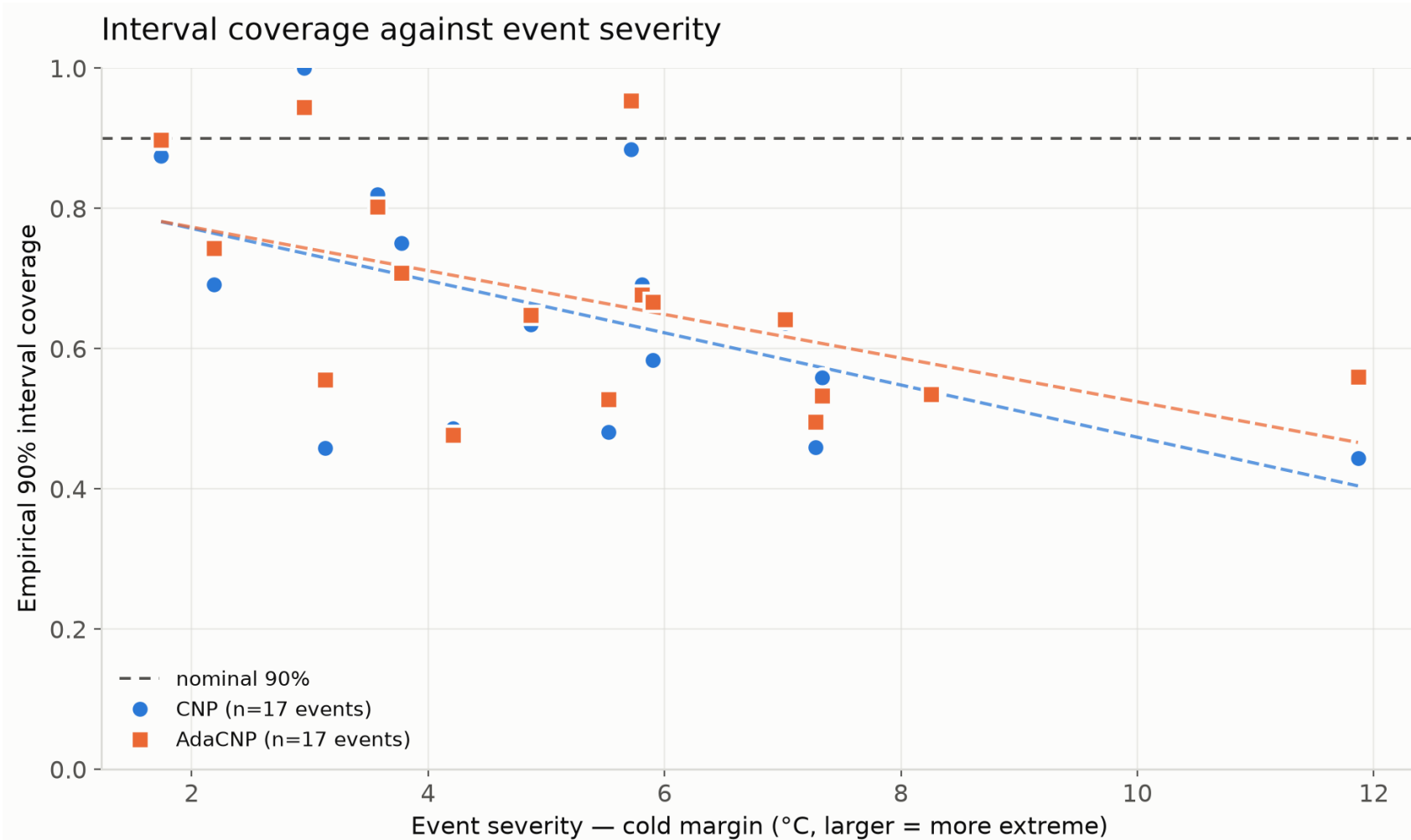
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Did any single event drive the result?



No. The sign holds across all 17 leave-one-out refits, the jackknife spans [+0.343, +0.501], and 11 of 17 events favor AdaCNP.

Does calibration get worse as events get worse?



CNP: slope -0.037/°C, $r=-0.55$, $p=0.02$ · AdaCNP: slope -0.031/°C, $r=-0.51$, $p=0.04$ — trend lines are descriptive, not a test

EXPLORATORY — outside the D-016 pre-registration